Hermitian duality of dihedral codes and application to quantum codes

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Abstract

Let D_n be the dihedral group of order $2n$, and let $\mathbb{F}_{q^2}$ be a finite field with q^2 elements, for a prime power q coprime with n. The goal of this note is to present a complete description of the hermitian dual of a D_n-code over $\mathbb{F}_{q^2}$ in terms of its expression as a direct sum of ideals within the Wedderburn-Artin decomposition of the group algebra $\mathbb{F}_{q^2}[D_n]$. As an application, a systematic method to build quantum error-correcting codes is illustrated, via the hermitian construction applied to self-orthogonal D_n-codes.

Key Words : linear codes, group algebras, dihedral groups, hermitian self-orthogonal codes, quantum codes

1 Introduction

Hereafter $\mathbb{F}_q$ will denote a finite field of q elements, for some prime power q, and n will be a positive integer coprime with n. Given a group G of order n, the $\mathbb{F}_q$-vector space $\mathbb{F}_q[G] := \left\{\sum_{g\in G} a_g g \mid a_g \in \mathbb{F}_q\right\}$, with basis the elements of G, and endowed with multiplication $\left(\sum_{g\in G} a_g g\right) \cdot \left(\sum_{g\in G} b_g g\right) := \sum_{g\in G}\left(\sum_{h\in G} a_h b_{h^{-1}g}\right) g$, is an algebra, called the *group algebra* of G over $\mathbb{F}_q$. A linear code $\mathcal{C} \subseteq \mathbb{F}_q^n$ is called a *(left) group code* (or simply a *G-code*) if

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there exists an isomorphism $\phi : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q[G]$ such that $\phi(\mathcal{C})$ is closed under (left) multiplication of every element of G. In other words, $\mathcal{C}$ is a G-code if $\phi(\mathcal{C})$ is a (left) ideal of $\mathbb{F}_q[G]$. Throughout this note all ideals are assumed to be left ideals.

The starting point of this algebraic approach to study linear codes may be placed in the papers [Ber67] and [Mac70], where S. Berman and F. MacWilliams independently introduced the notion of a group code, respectively. In fact, it was discovered in [Ber67] that Reed-Muller codes over a finite field of characteristic 2 may be seen as ideals in the group algebra of an elementary abelian 2-group, and this was later extended for Generalised Reed-Muller codes (*i.e.* for odd characteristic) in [Cha82]. Several families of linear codes have been proved to be G-codes for certain finite groups G, and we refer the interested reader to the survey [KS01].

The main interest in the study of group codes is that their algebraic structure can be deeply analysed using powerful tools from representation theory, some of which will be described later. An instance of this feature is given in [LM92], which takes advantage of the underlying algebra structure to propose a new decoding algorithm for Reed-Muller codes. This illustrates once again that the more mathematical structure an object possesses, the better are the descriptions we can obtain.

Abelian group codes have been extensively studied. However, the research on non-abelian group codes is gaining an increasing interest in recent years (*cf.* [GYW20, GY21, VD21], to name a few). One of the main motivations for this is that quantum computers seem not able to break, so far, the security of public-key cryptosystems based on non-abelian groups codes (see [Sen17]). We emphasize that a G-code can also be an H-code, for another group H that is distinct from G; indeed, it may even be the case that G is abelian and H is not. In any case, as it is mentioned in the expository paper [GMMM19], non-abelian group codes that cannot be realised as G-codes for some abelian group G have length at least 24, and dimension at least 4.

Brochero-Martínez described in [BM15] the Wedderburn-Artin decomposition of the group algebra $\mathbb{F}_q[D_n]$ of any dihedral group of order $2n$ in the semisimple case, *i.e.* whenever the characteristic of the field $\mathbb{F}_q$ is coprime with $2n$. Later on, the decomposition of $\mathbb{F}_q[D_n]$ was utilised by Vedenev and Deundyak to obtain in [VD21] a detailed algebraic description of all D_n-codes and their euclidean dual codes. In [SCSES26a] we applied their research to build quantum CSS dihedral codes, alghough we were not able to break any record regarding the parameters of the best known quantum codes; but on the upside, our systematic approach avoided brute force computations to check whether the considered pair of D_n-codes were nested (as it is done, for instance, in [YZ24]). Thus one of our main objectives was to adapt this systematic approach to the hermitian construction, and so we needed to produce first hermitian self-orthogonal D_n-codes over $\mathbb{F}_{q^2}$ in terms of the decomposition of $\mathbb{F}_{q^2}[D_n]$. It is worth to remark that the hermitian construction of quantum codes (see Theorem 2.2 below) often yields better parameters than the (euclidean) CSS construction in experimental results, since it involves larger alphabets.

This extended abstract is part of a paper in progress (*cf.* [SCSES26b]) where we have carried out this previous idea. More concretely, we refine the Wedderburn-Artin decomposition given in [BM15] of $\mathbb{F}_{q^2}[D_n]$ to provide a full characterisation of the hermitian dual code of any D_n-code over $\mathbb{F}_{q^2}$ whenever the characteristic of the field is coprime with n. Cao *et al.* also determined algebraic expressions of D_n-codes, their euclidean dual codes and their hermitian dual codes but with a different approach, namely the concatenated structure of dihedral codes (see [CCM22, CCF23]). However, our alternative approach via the group algebra structure reduces considerably the laborious computations given in those papers, and it also serves to detect some innacuracies there. As a consequence of our theoretical results, we provide numerical examples of quantum error-correcting codes that can be constructed from hermitian self-orthogonal dihedral codes and which reach the best known parameters. We also remark that the authors of [WBHI25] also obtained dihedral quantum codes, but making use of the lifted product construction; and the quantum codes displayed in that paper have good code rate.

2 Preliminaries

All groups considered hereafter are supposed to be finite. Let us denote by $\mathcal{M}_n(R)$ the ring of $(n \times n)$-matrices over a ring R, and by $\langle X \rangle$ the ideal of R generated by a non-empty subset X of R. The rest of the notation not explained here is standard in the context of group theory or coding theory.

2.1 On group codes

The group algebra $\mathbb{F}_q[G]$ is said to be *semisimple* if it can be realised as a direct sum of simple $\mathbb{F}_q[G]$-modules, and this occurs, by Maschke's theorem, if and only if the characteristic of the field does not divide the order of G (*cf.* [DH92]). In this case, every ideal of $\mathbb{F}_q[G]$ is always a direct summand and thus it must be principal. Moreover, the well-known Wedderburn-Artin theorem ([DH92, Theorem 4.4, p. 112]) ensures that when $\mathbb{F}_q[G]$ is semisimple it can be realised as a direct sum of matrix algebras:

Theorem 2.1 (Wedderburn-Artin decomposition for finite group algebras)**.** *Let G be a finite group such that $\mathbb{F}_q[G]$ is a semisimple group algebra. Then $\mathbb{F}_q[G]$ is isomorphic, as $\mathbb{F}_q$-algebra, to the direct sum of some matrix rings over suitable extensions of $\mathbb{F}_q$. Specifically, one has:*

$$\mathbb{F}_q[G] \cong \bigoplus_{i=1}^{s} \mathcal{M}_{n_i}\left(\mathbb{F}_{q^{r_i}}\right)$$

satisfying $|G| = \sum_{i=1}^{s} n_i^2 r_i$.

Let $\mathcal{C}$ be a G-code, and let $I_{\mathcal{C}}$ be the corresponding ideal in $\mathbb{F}_q[G]$ mentioned at the beginning of the Introduction. Notice that when $\mathbb{F}_q[G]$ is semisimple, by the previous theorem, we can always write $I_{\mathcal{C}}$ as a (unique) sum $I_1 \oplus \cdots \oplus I_s$, where I_i is an ideal of $\mathcal{M}_{n_i}(\mathbb{F}_{q^{r_i}})$ for every $1 \leq i \leq s$. Moreover, for every $1 \leq i \leq s$, each $\mathcal{M}_{n_i}(\mathbb{F}_{q^{r_i}})$ is a principal ideal ring, so each ideal I_i is generated by a unique matrix M_i in row reduced echelon form, that is,

$$I_i = \langle M_i \rangle = \{XM_i \mid X \in \mathcal{M}_{n_i}(\mathbb{F}_{q^{r_i}})\}.$$

In particular, the dimension of each ideal I_i as $\mathbb{F}_q$-vector space is just $n_i r_i \operatorname{rank}(M_i)$, and consequently the sum of these dimensions yields the dimension of the group code $\mathcal{C}$ (see [SCSES26a, Lemma 3.7, Theorem 3.8]), that is:

$$\dim_{\mathbb{F}_q}(\mathcal{C}) = \sum_{i=1}^{s} n_i r_i \operatorname{rank}(M_i). \tag{1}$$

This approach is fundamental, since it allows us to give all possible G-codes and their dimensions whenever the Wedderburn-Artin decomposition of $\mathbb{F}_q[G]$ is known.

Given two elements $x = (x_1, x_2, ..., x_n)$ and $y = (y_1, y_2, ..., y_n)$ in $\mathbb{F}_{q^2}^n$, their *euclidean inner-product* is defined as $(x \mid y)_e := \sum_{i=1}^{n} x_i y_i$, while the *hermitian inner-product* is $(x \mid y)_h := \sum_{i=1}^{n} x_i y_i^q$. Thus, the *euclidean dual code* and the *hermitian dual code* of a linear code $\mathcal{C}$ of $\mathbb{F}_{q^2}^n$ are, respectively,

$$\mathcal{C}^{\perp_e} := \{y \in \mathbb{F}_{q^2}^n \ : \ (x|y)_e = 0, \ \forall\, x \in \mathcal{C}\},$$

and

$$\mathcal{C}^{\perp_h} := \{y \in \mathbb{F}_{q^2}^n \ : \ (x|y)_h = 0, \ \forall\, x \in \mathcal{C}\}.$$

Moreover, it is easy to see that

$$\mathcal{C}^{\perp_h} = (\mathcal{C}^{\perp_e})^q := \{(y_1^q, ..., y_n^q) \mid (y_1, ..., y_n) \in C^{\perp_e}\}.$$

2.2 On quantum codes

A q-ary *quantum error-correcting code* (QECC) $\mathcal{Q}$ of length n is a subspace of the n-fold tensor product $(\mathbb{C}^q)^{\otimes n}$ of the complex vector space $\mathbb{C}^q$. The dimension of the code $\mathcal{Q}$ is therefore $\dim_{\mathbb{C}}(\mathcal{Q}) \leq q^n$. This kind of codes are useful to reduce the effects of environmental and operational noise in quantum information processing. Here we focus on the so-called *stabiliser codes* based on linear codes over finite fields and, within them, we concretely deal with *quantum CSS codes*, whose name is due to Calderbank, Shor and Steane. In this case, the dimension is q^k for some positive integer $k \leq n$, and we write its parameters as $[[n, k, d]]_q$, where the minimum distance d

means that any error acting on at most $d-1$ positions of the tensor factors (the so-called *qubits*) can be detected or has no effect on the code. The basic theory of this kind of codes can be found on [KKKS06], and for more information on general quantum codes, see, for example, [Gua20].

In the following theorem we present the hermitian construction of quantum codes. Below $\mathrm{wgt}(c)$ denotes the Hamming weight of a codeword c in a linear code $\mathcal{C}$. Recall that $\mathcal{C}$ is called *hermitian self-orthogonal* if $\mathcal{C} \subseteq \mathcal{C}^{\perp_h}$.

Theorem 2.2. ([CRSS98]) *If $\mathcal{C} \subseteq \mathbb{F}_{q^2}^n$ is an hermitian self-orthogonal linear code with parameters $[n,k,d]_{q^2}$, then there exists an $[[n, n-2k, d_{\mathcal{Q}}]]_q$ quantum stabiliser code $\mathcal{Q}$ with $d_{\mathcal{Q}} := d$ if $\mathcal{C} = \mathcal{C}^{\perp_h}$, and otherwise*

$$d_{\mathcal{Q}} := \min\{\mathrm{wgt}(x) \mid x \in \mathcal{C}^{\perp_h} \setminus \mathcal{C}\} \geq d.$$

3 A suitable decomposition of $\mathbb{F}_{q^2}[D_n]$

When the group algebra $\mathbb{F}_q[G]$ is semisimple, the decomposition provided by the Wedderburn-Artin theorem (Thm. 2.1) has been concretised in several cases. With regard to dihedral groups, such a decomposition is given in [BM15, Theorem 3.1] and we are going to rewrite it appropiately to address the study of the hermitian dual of any dihedral code over $\mathbb{F}_{q^2}$.

Given a polynomial $f = \sum_{i=0}^m a_i \mathrm{x}^i \in \mathbb{F}_{q^2}[\mathrm{x}]$, we denote by $\overline{f} := \sum_{i=0}^m a_i^q \mathrm{x}^i$ its *conjugate polynomial.* In particular, f is called *self-conjugate* whenever $f = \overline{f}$. We denote by f^* the *reciprocal* polynomial of f, *i.e.* $f^*(\mathrm{x}) = f(0)^{-1}\mathrm{x}^{deg(f)} f(\mathrm{x}^{-1})$. The polynomial f is said to be *self-reciprocal* if $f = f^*$. We write $f^\dagger$ for the conjugate of the reciprocal polynomial of f, that is, $f^\dagger = \overline{f^*}$.

Notation 3.1. *Suppose that $(q,n) = 1$, and decompose the polynomial $\mathrm{x}^n - 1$ into irreducible factors in $\mathbb{F}_{q^2}[\mathrm{x}]$ as follows:*

$$\mathrm{x}^n - 1 = \prod_{j \in J_0} f_j \prod_{j \in J_1} f_j \overline{f_j} \prod_{j \in J_2 \cup J_3} f_j f_j^* \prod_{j \in J_4} f_j f_j^* \overline{f_j} f_j^\dagger, \tag{2}$$

where:

- *$j \in J_0$ if and only if $\overline{f_j} = f_j = f_j^*$. In this case, $f_j \in \{\mathrm{x}-1, \mathrm{x}+1\}$.*
- *$j \in J_1$ if and only if $\overline{f_j} \neq f_j = f_j^*$;*
- *$j \in J_2$ if and only if $\overline{f_j} = f_j \neq f_j^*$;*
- *$j \in J_3$ if and only if $\overline{f_j} = f_j^* \neq f_j$;*
- *$j \in J_4$ if and only if $f_j, \overline{f_j}$ and f_j^* are all different.*

Moreover, set $J := \bigcup_{i=0}^{4} J_i$, α_j for a root of f_j, and $r_j := \deg(f_j)$.

Given the group algebra $\mathbb{F}_{q^2}[D_n]$, let us restate the isomorphism given in [BM15, Theorem 3.1] by considering the decomposition (2) and by grouping together those blocks associated to pairs of conjugate polynomials. Below C_2 denotes the cyclic group of order 2 generated by the matrix $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$.

Theorem 3.2. *Following Notation 3.1, there exists an isomorphism ρ of $\mathbb{F}_{q^2}$-algebras*

$$\mathbb{F}_{q^2}[D_n] \overset{\rho}{\cong} \bigoplus_{j \in J} A_j,$$

where A_j is equal to:

- $\mathbb{F}_{q^2} \oplus \mathbb{F}_{q^2}$, *if $j \in J_0$ and $2 \nmid q$.*
- $\mathbb{F}_{q^2}[C_2]$, *if $j \in J_0$ and $2 \mid q$.*
- $\mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j + \alpha_j^{-1})\right) \oplus \mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j^q + \alpha_j^{-q})\right)$, *if $j \in J_1$.*
- $\mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j)\right)$, *if $j \in J_2 \cup J_3$.*
- $\mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j)\right) \oplus \mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j^q)\right)$, *if $j \in J_4$.*

4 Main results

The aim of this section is to present a full description of the hermitian dual of a dihedral code based on its expression as a direct sum of ideals within the Wedderburn-Artin decomposition of $\mathbb{F}_{q^2}[D_n]$ provided in Theorem 3.2.

Consider the automorphism $\sigma : x \mapsto x^q$ of $\mathbb{F}_{q^2}$. As mentioned in Subsection 2.1, it is easy to see that, if $\mathcal{C} \subseteq \mathbb{F}_{q^2}$ is a linear code, then $\mathcal{C}^{\perp_h}$ can be computed from the euclidean duality $\mathcal{C}^{\perp_e}$ by simply applying σ to every component of each codeword of $\mathcal{C}^{\perp_e}$. Taking into account this observation, and the description of the euclidean dual of any dihedral code already given in [VD21], our goal is then to find φ such that the following diagram is commutative. Below conj_q denotes the natural linear extension to $\mathbb{F}_{q^2}[G]$ of the automorphism $\sigma : x \mapsto x^q$ of $\mathbb{F}_{q^2}$.

$$\begin{tikzcd} \mathbb{F}_{q^2}[D_n] \arrow[r, "\operatorname{conj}_q"] \arrow[d, "\rho"'] & \mathbb{F}_{q^2}[D_n] \arrow[d, "\rho"] \\ \bigoplus_{j \in J} A_j \arrow[r, "\varphi"'] & \bigoplus_{j \in J} A_j \end{tikzcd} \tag{3}$$

In this way, if $I_{\mathcal{C}^{\perp_h}}$ is the ideal corresponding to $\mathcal{C}^{\perp_h}$, then we can find $\rho(I_{\mathcal{C}^{\perp_h}})$ through φ:

$$\rho(I_{\mathcal{C}^{\perp_h}}) = (\rho \circ \operatorname{conj}_q)(I_{\mathcal{C}^{\perp_e}}) = \varphi(\rho(I_{\mathcal{C}^{\perp_e}})).$$

Theorem 4.1. *Following Notation 3.1, the diagram (3) is commutative if*

$$\varphi = \bigoplus_{j\in J} \varphi_j, \ \text{where } \varphi_j : A_j \to A_j \text{ is given by}$$

- $\varphi_j(x,y) = (x^q, y^q)$, *if* $j \in J_0$ *and* $2 \nmid q$;
- $\varphi_j(X) = \mathcal{H}^1(X)$, *if* $j \in J_0$ *and* $2 \mid q$;
- $\varphi_j(X,Y) = \Big(\sigma_j \circ \mathcal{H}^{2r_j-1} \circ \overline{\sigma_j}^{-1}(Y), \overline{\sigma_j} \circ \mathcal{H}^1 \circ \sigma_j^{-1}(X)\Big)$, *if* $j \in J_1$;
- $\varphi_j(X) = \mathcal{H}^{r_j}(X)$, *if* $j \in J_2$;
- $\varphi_j(X) = \mathcal{T}^{r_j}(X)$, *if* $j \in J_3$;
- $\varphi_j(X,Y) = \big(\mathcal{H}^{2r_j-1}(Y), \mathcal{H}^1(X)\big)$, *if* $j \in J_4$,

where $\sigma_j(X) := Z_j^{-1} X Z_j$ *with* $Z_j := \begin{bmatrix} 1 & -\alpha_j \\ 1 & -\alpha_j^{-1} \end{bmatrix}$, $\overline{\sigma_j}(X) := \overline{Z_j}^{-1} X \overline{Z_j}$ *with* $\overline{Z_j} := \begin{bmatrix} 1 & -\alpha_j^q \\ 1 & -\alpha_j^{-q} \end{bmatrix}$, *and* $\mathcal{H}^m$ *and* $\mathcal{T}^m$ *are the following maps:*

$$\mathcal{H}^m : \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a^{q^m} & b^{q^m} \\ c^{q^m} & d^{q^m} \end{bmatrix}, \quad \mathcal{T}^m : \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} d^{q^m} & c^{q^m} \\ b^{q^m} & a^{q^m} \end{bmatrix}.$$

Using the previous theorem, we demonstrate the next result, which gives us the complete description of the hermitian dual of a dihedral code.

Theorem 4.2. *Following Notation 3.1, let* $I_{\mathcal{C}} \subseteq \mathbb{F}_{q^2}[D_n]$ *be the ideal corresponding to a dihedral code* $\mathcal{C}$ *over* $\mathbb{F}_{q^2}$, *which can be expressed by Theorem 3.2 as*

$$I_{\mathcal{C}} \cong \bigoplus_{j\in J} \mathcal{I}_j \subseteq \bigoplus_{j\in J} A_j,$$

where $\mathcal{I}_j$ *is an ideal of* A_j. *Set* $\mathcal{I}_j = \langle X\rangle \oplus \langle Y\rangle$ *for* $j \in J_1 \cup J_4$, *and* $\mathcal{I}_j = \langle X\rangle$ *for* $j \in J_2 \cup J_3$, *with* X, Y *matrices in row reduced echelon form. Then the ideal corresponding to the hermitian dual code* $\mathcal{C}^{\perp_h}$ *is*

$$I_{\mathcal{C}^{\perp_h}} \cong \bigoplus_{j\in J} \widehat{\mathcal{I}}_j \subseteq \bigoplus_{j\in J} A_j,$$

where:
for $j \in J_0$ *and* $2 \nmid q$*:*

$$\widehat{\mathcal{I}}_j = \begin{cases} \mathbb{F}_{q^2} \oplus \mathbb{F}_{q^2} & \text{if } \mathcal{I}_j = \mathbf{0} \oplus \mathbf{0} \\ \mathbf{0} \oplus \mathbb{F}_{q^2} & \text{if } \mathcal{I}_j = \mathbb{F}_{q^2} \oplus \mathbf{0} \\ \mathbb{F}_{q^2} \oplus \mathbf{0} & \text{if } \mathcal{I}_j = \mathbf{0} \oplus \mathbb{F}_{q^2} \\ \mathbf{0} \oplus \mathbf{0} & \text{if } \mathcal{I}_j = \mathbb{F}_{q^2} \oplus \mathbb{F}_{q^2} \end{cases}$$

for $j \in J_0$ and $2 \mid q$:

$$\widehat{\mathcal{I}}_j = \begin{cases} \mathbb{F}_{q^2}[C_2] & \text{if } \mathcal{I}_j = \mathbf{0} \\ \left\langle \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle & \text{if } \mathcal{I}_j = \left\langle \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \\ \mathbf{0} & \text{if } \mathcal{I}_j = \mathbb{F}_{q^2}[C_2] \end{cases}$$

for $j \in J_1$: $\widehat{\mathcal{I}}_j = \langle Y_1 \rangle \oplus \langle X_1 \rangle$, and one of the following holds for X_1 and Y_1:

- $X_1 = I - X$, *if* $X = \mathbf{0}$ *or* $X = I$
- $X_1 = \begin{bmatrix} 2 & -(\alpha_j + \alpha_j^{-1})^q \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
- $X_1 = \begin{bmatrix} (2\lambda_j + \alpha_j + \alpha_j^{-1})^q & (-2 - (\alpha_j + \alpha_j^{-1})\lambda_j)^q \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix}$ *with* $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j + \alpha_j^{-1})$
- $Y_1 = I - Y$, *if* $Y = \mathbf{0}$ *or* $Y = I$
- $Y_1 = \begin{bmatrix} 2 & -(\alpha_j + \alpha_j^{-1}) \\ 0 & 0 \end{bmatrix}$, *if* $Y = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
- $Y_1 = \begin{bmatrix} 2\lambda_j^{q^{2r_j - 1}} + \alpha_j + \alpha_j^{-1} & -2 - (\alpha_j + \alpha_j^{-1})\lambda_j^{q^{2r_j - 1}} \\ 0 & 0 \end{bmatrix}$, *if* $Y = \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix}$ *with* $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j^q + \alpha_j^{-q})$

for $j \in J_2$: $\widehat{\mathcal{I}}_j = \langle X_1 \rangle$, and one of the following holds:

- $X_1 = I - X$, *if* $X = \mathbf{0}$ *or* $X = I$
- $X_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
- $X_1 = \begin{bmatrix} 1 & -\lambda_j^{q^{r_j}} \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix}$ *with* $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j)$

for $j \in J_3$: $\widehat{\mathcal{I}}_j = \langle X_1 \rangle$, and one of the following holds:

- $X_1 = I - X$, *if* $X = \mathbf{0}$ *or* $X = I$
- $X_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
- $X_1 = \begin{bmatrix} -\lambda_j^{q^{r_j}} & 1 \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix}$ *with* $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j)$

for $j \in J_4$: $\widehat{\mathcal{I}}_j = \langle Y_1 \rangle \oplus \langle X_1 \rangle$, *and one of the following holds for* X_1 *and* Y_1:

- $X_1 = I - X$, *if* $X = \mathbf{0}$ *or* $X = I$
- $X_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
- $X_1 = \begin{bmatrix} 1 & -\lambda_j^q \\ 0 & 0 \end{bmatrix}$, *if* $X = \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix}$ *with* $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j)$
- $Y_1 = I - Y$, *if* $Y = \mathbf{0}$ *or* $Y = I$
- $Y_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, *if* $Y = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
- $Y_1 = \begin{bmatrix} 1 & -\lambda_j^{q^{2r_j-1}} \\ 0 & 0 \end{bmatrix}$, *if* $Y = \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix}$ *with* $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j^q)$

Corollary 4.3. *Following the notation of the previous theorem,* $\mathcal{C} \subseteq \mathcal{C}^{\perp_h}$ *if and only if the following conditions are satisfied:*

- *If* $j \in J_0$ *and* q *is odd, then* $\mathcal{I}_j = \mathbf{0} \oplus \mathbf{0}$.
- *If* $j \in J_0$ *and* q *is even, then* $\mathcal{I}_j = \mathbf{0} \oplus \mathbf{0}$ *or* $\mathcal{I}_j = \langle [\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}] \rangle$.
- *If* $j \in J_1$, *then either one of the summands of* $\mathcal{I}_j$ *is* $\mathbf{0}$, *or* $\mathcal{I}_j$ *has the form:*
  - ▶ $\left\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 2 & -(\alpha_j^q + \alpha_j^{-q}) \\ 0 & 0 \end{bmatrix} \right\rangle$,
  - ▶ $\left\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} (2\lambda_j + \alpha_j + \alpha_j^{-1})^q & [-2 - \lambda_j(\alpha_j + \alpha_j^{-1})]^q) \\ 0 & 0 \end{bmatrix} \right\rangle$.
- *If* $j \in J_2$, *then* $\mathcal{I}_j$ *is either* $\mathbf{0}$, $\langle [\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}] \rangle$, *or* $\left\langle \left[\begin{smallmatrix} 1 & \lambda_j \\ 0 & 0 \end{smallmatrix}\right] \right\rangle$ *with* $\lambda_j = -\lambda_j^{q^{r_j}}$.
- *If* $j \in J_3$, *then* $\mathcal{I}_j$ *is either* $\mathbf{0}$, *or* $\left\langle \left[\begin{smallmatrix} 1 & \lambda_j \\ 0 & 0 \end{smallmatrix}\right] \right\rangle$ *with* $0 \neq \lambda_j = -\lambda_j^{-q^{r_j}}$.
- *If* $j \in J_4$, *then either one of the summands of* $\mathcal{I}_j$ *is* $\mathbf{0}$, *or* $\mathcal{I}_j$ *has the form:*
  - ▶ $\left\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle$,
  - ▶ $\left\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & -\lambda_j^q \\ 0 & 0 \end{bmatrix} \right\rangle$.

In particular, the number of hermitian self-orthogonal D_n*-codes over* $\mathbb{F}_{q^2}$ *is*

$$\epsilon \prod_{j \in J_1} (3q^{r_j} + 6) \prod_{j \in J_2 \cup J_3} (q^{r_j} + 2) \prod_{j \in J_4} \left(3q^{2r_j} + 6\right),$$

with $\epsilon = 1$ *if* q *is odd and* $\epsilon = 2$ *otherwise.*

In view of the last assertion, the computation of the number of hermitian self-orthogonal D_n-codes over $\mathbb{F}_{q^2}$ in [CCF23, Theorem 4.3] does not seem to be correct.

Example 4.4. *Let $q = 9$ and D_{16} the dihedral group of order 32. Let ω be a primitive element of $\mathbb{F}_9$ and ν a primitive element of $\mathbb{F}_{9^2}$. The polynomial $\mathrm{x}^{16} - 1$ factorises in $\mathbb{F}_9[\mathrm{x}]$ as:*

$$\begin{aligned}\mathrm{x}^{16} - 1 = &(\mathrm{x} - 1)(\mathrm{x} + 1)(\mathrm{x} + \omega)(\mathrm{x} + \omega^2)(\mathrm{x} + \omega^3)(\mathrm{x} + \omega^5)\\ &(\mathrm{x} + \omega^6)(\mathrm{x} + \omega^7)(\mathrm{x}^2 + \omega)(\mathrm{x}^2 + \omega^3)(\mathrm{x}^2 + \omega^5)(\mathrm{x}^2 + \omega^7).\end{aligned}$$

Therefore, following Notation 3.1, one has $f_1 = \mathrm{x}-1$, $f_2 = \mathrm{x}+1$, $f_3 = \mathrm{x}+\omega$, $f_4 = \mathrm{x} + \omega^2$, $\bar{f}_3 = \mathrm{x} + \omega^3$, $\overline{f_3^*} = \mathrm{x} + \omega^5$, $f_4^* = \bar{f}_4 = \mathrm{x} + \omega^6$, $f_3^* = \mathrm{x} + \omega^7$, $f_3 = \mathrm{x}^2 + \omega$, $\bar{f}_5 = \mathrm{x}^2 + \omega^3$, $\overline{f_5^*} = \mathrm{x}^2 + \omega^5$ and $f_5^* = \mathrm{x}^2 + \omega^7$. It follows $J_0 = \{1, 2\}$, $J_1 = J_2 = \emptyset$, $J_3 = \{4\}$ and $J_4 = \{3, 5\}$. By Theorem 3.2 the group algebra $\mathbb{F}_9[D_{16}]$ decomposes in the following way:

$$\begin{aligned}\mathbb{F}_9[D_{16}] \cong &(\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus (\mathcal{M}_2(\mathbb{F}_9) \oplus \mathcal{M}_2(\mathbb{F}_9))\\ &\oplus \mathcal{M}_2(\mathbb{F}_9) \oplus (\mathcal{M}_2(\mathbb{F}_{9^2}) \oplus \mathcal{M}_2(\mathbb{F}_{9^2})).\end{aligned}$$

Consider now the dihedral code $\mathcal{C}$ over $\mathbb{F}_9$ such that

$$\begin{aligned}I_{\mathcal{C}} \cong &(\mathbf{0} \oplus \mathbf{0}) \oplus (\mathbf{0} \oplus \mathbf{0}) \oplus \left\langle \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle\\ &\oplus \left\langle \begin{bmatrix} 1 & \omega^7 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^{14} \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^2 \\ 0 & 0 \end{bmatrix} \right\rangle.\end{aligned}$$

The dimension of $\mathcal{C}$ can be easily computed via (1), and indeed it is 12. By Corollary 4.3 we can affirm that it is hermitian self-orthogonal, and in fact by Theorem 4.2 we have:

$$\begin{aligned}I_{\mathcal{C}^{\perp_h}} \cong &(\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus \left\langle \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \mathcal{M}_2(\mathbb{F}_9)\\ &\oplus \left\langle \begin{bmatrix} 1 & \omega^7 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^{14} \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^2 \\ 0 & 0 \end{bmatrix} \right\rangle.\end{aligned}$$

Consequently, Theorem 2.2 ensures the existence of a QECC of length 32 and dimension 8, and with Magma *we computed its minimum distance, which is 8. Hence we have obtained a QECC that reaches the best known parameters $[[32, 8, 8]]_3$, according to Grassl's tables [Gra07].*

This strategy has also been applied for other combinations of q and n, and we have produced, for instance, QECCs with (best known, according to [Gra07]) parameters $[[32, 16, 6]]_3$, $[[34, 10, 7]]_2$ and $[[42, 12, 8]]_2$.

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